

SR NEXT - Instagram Influencer Analytics Report

1. Executive Summary

This report outlines the methodology, tools, and outcomes for the SR NEXT Self-Guided Internship project. The project focuses on collecting, analyzing, and ranking Instagram influencers with under 1 million followers using Python, AI, and Machine Learning techniques. The project was successfully divided into two core phases: Data Collection and AI-Powered Analysis.

2. Phase 1: Data Collection & Preprocessing

Objective: To collect public Instagram data and prepare a structured influencer dataset.

Data Sources & Criteria:

- Target: Instagram influencers and pages with less than 1M followers.
- Dataset Size: Targeted over 2000 profiles.
- Key Data Points: Name, Username, Followers, Engagement Rate, Contact Details, Posting Frequency, Hashtags, and Niche/Category.

Tools Used:

- Python, Pandas, BeautifulSoup, Selenium, and standard requests modules.

Process:

Data was collected by identifying target profiles based on follower count. Publicly available metrics were scraped, cleaned, and compiled into a structured CSV format. Missing values and outliers were handled during the preprocessing step to ensure data quality and integrity for the next phase.

3. Phase 2: AI-Powered Analysis & Ranking

Objective: To analyze collected data, apply AI/ML techniques, and rank influencers.

Feature Engineering:

- Calculated Engagement Rate $((\text{Likes} + \text{Comments}) / \text{Followers})$ based on the latest posts.
- Extracted hashtag diversity and categorized content types.

AI/ML Models:

- NLP: Utilized for caption analysis and content classification to categorize influencers (e.g., Tech, Fashion).
- Predictive Modeling: Employed basic ML models using scikit-learn to predict engagement trends and the

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likelihood of automation tool adoption based on posting patterns.

Ranking Methodology:

- A composite score was created by weighting engagement metrics and AI predictions. This score was used to generate the final ranked influencer list.

Visualization:

- A Streamlit/Plotly dashboard was developed to provide interactive visual insights into the top influencers per category.

4. Conclusion & Deliverables

The project successfully demonstrated an end-to-end data pipeline: from web scraping and data cleaning to machine learning and data visualization. The final deliverables include:

1. A comprehensive, cleaned dataset (CSV) of influencers.
2. A final ranked list of influencers based on AI predictions.
3. An interactive visualization dashboard.